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				Date Prep.		15.11.2018		PROJECT TITLE				Scale				= H05			
				Prepared		YR		MV Switchgear for MNA Serang Plant				PT. ABB Sakti Industri		Customer		PT Multimas Nabati Asahan		& 308	
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UniSec

Medium Voltage Switchgear

PT MULTIMAS NABATI ASAHAN

MVD-31 20kV Texturing Plant

SCHEMATIC DIAGRAM

DRC BUS PT

H05

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					Prepared	YR	MV Switchgear for MNA Serang Plant			MVD-31 20kV Texturing Plant		Customer	PT Multimas Nabati Asahan	& 308	
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A	-BGI31	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION						-CA	CAPACITORS						A									
	-BGI32	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION						-CA1	CAPACITOR FOR FRONT VENTILATOR															
	-BGI33	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)						-CA2	CAPACITOR FOR REAR VENTILATOR															
	-BGI41...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION						-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.															
B	-BGI42	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION						-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT						B									
	-BGI43...	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION						-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE															
	-BGI42	POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION						-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTMENT															
	-BGI51	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION						-EB1	HEATER LOCATED IN CABLE COMPARTMENT															
C	-BGI52	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION						-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT						C									
	-BGI53	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION						-EB3	HEATER LOCATED IN MOTOR CABINET															
	-BGK	POSITION SWITCH OF KEY LOCK						-EB4	HEATER FOR VENTILATION TEST															
	-BGL1	POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1						-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.															
D	-BGS1...-BGS4	POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS						-EB7	HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE						D									
	-BGS5	PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION						-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE															
	-BGT1	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION						-FA1	SURGE ARRESTER LOCATED ON PHASE L1															
	-BGT2	POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION						-FA2	SURGE ARRESTER LOCATED ON PHASE L2															
E	-BGT3	POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION						-FA3	SURGE ARRESTER LOCATED ON PHASE L3						E									
	-BGT4	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION						-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS															
	-BGT5	POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION						-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1															
	-BGT6	POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION						-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2															
F	-BGT7	POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION						-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3						F									
	-BGT8	POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION						-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER															
	-BGT11	PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION						-FCM2...-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS															
	-BGT12	PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION						-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1															
	-BGV1	POSITION SWITCH OF FRONT VENTILATOR						-FCM12	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2															
	-BGV2	POSITION SWITCH OF REAR VENTILATOR						-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3															
	-BM	HYGROSTAT						-FCT	THERMAL OVERLOAD RELEASES															
	-BPA4	PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GA	GENERATORS															
	-BPA5	PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ1	FRONT VENTILATOR															
	-BPA6	PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ2	REAR VENTILATOR															
	-BPA7	PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK															
	-BPS1	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER						-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE															
	-BPS2	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2						-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE															
	-BPS3	PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3						-KFA3...-KFA9	AUXILIARY RELAYS OR CONTACTORS															
	-BR	FLAME DETECTORS, SMOKE DETECTORS						-KFC	CLOSING RELAYS OR CONTACTORS															
	-BT	THERMOSTAT						-KFI	INTEGRATED CIRCUITS															
	-BT1	THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS						-KFL	LOCKOUT RELAY															
	-BT2	THERMOSTAT FOR ENERGIZATION OF VENTILATORS						-KFM	MICROPROCESSORS															
	-BT3	THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM						-KFN	ANTIPUMPING RELAYS															
	-BU	BUCHHOLZ RELAY						-KFO	OPENING RELAYS OR CONTACTORS															
	-BX1	UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING						-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)															
	-BX2	ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING						-KFR	RECLOSING RELAYS															
	-BZ,-BZ4	MULTIFUNCTION SENSORS						-KFS	SYNCHRONIZING DEVICES															
	-BZ1	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1						-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS															
	-BZ2	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2						-KFZ	CONTROLLERS															
	-BZ3	COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3						-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER															
								-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD															
								-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1															
								-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2															
								-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS															
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A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE	-QBH	MANUAL CIRCUIT-BREAKERS
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2	-QBL	LINKS
	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS	-QBM	MINIATURE SWITCH-DISCONNECTORS
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING	-QBS	SWITCH-DISCONNECTORS
B	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT	-QBT	TRUCK
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER	-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING
	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS	-QCE	EARTHING SWITCH
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER	-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)
	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER	-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER	-QQ	CLUTCHES
B	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS	-RAA	ANTIFERRORESONANCE RESISTOR
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER	-RAD	DIODES
	-PFB	BLUE SIGNAL LAMPS	-RAI	INDUCTORS
	-PFF	FLAG RELAYS	-RAR	RESISTORS
	-PFG	GREEN SIGNAL LAMPS	-RF	FILTERS
	-PFR	RED SIGNAL LAMPS	-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING
C	-PFS	SHORT CIRCUIT INDICATORS	-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT
	-PFV	VOLTAGE INDICATORS	-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE
	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE	-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE	-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD
C	-PFW	WHITE SIGNAL LAMPS	-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS	-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2
	-PFY	YELLOW SIGNAL LAMPS	-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2
	-PGA	AMMETERS	-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS
	-PGC	COUNTERS	-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4
	-PGF	FREQUENCYMETERS	-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5
D	-PGH	HOURMETERS	-SFA	AMMETRIC SWITCHES
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE	-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS
	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)	-SFL	LOCKING CONTACTS
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)	-SFO	OPENING PUSH-BUTTONS
	-PGJ	ACTIVE ENERGY METERS	-SFR	RESET PUSH-BUTTONS
	-PGK	REACTIVE ENERGY METERS	-SFS	SELECTOR SWITCHES
E	-PGM	MULTIFUNCTION INDICATORS	-SFT	TEST PUSH-BUTTONS
	-PGP	POWER-FACTOR METERS	-SFU	UNLOCKING PUSH-BUTTONS
	-PGQ	VARMETERS	-SFV	VOLTMETRIC SWITCHES
	-PGS	SYNCHRONOSCOPES	-TA	POWER TRANSFORMERS
	-PGV	VOLTMETERS	-TB	CONVERTER
	-PGW	WATTMETERS	-TB1	RECTIFIER FOR FIRST OPENING RELEASE
E	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)	-TB2	RECTIFIER FOR SECOND OPENING RELEASE
	-QAB	CIRCUIT-BREAKERS	-TB3	RECTIFIER FOR CLOSING RELEASE
	-QAC	CONTACTORS (FOR POWER)	-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING
	-QBD	DISCONNECTORS	-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT
	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)	-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)	-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER
	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)		
	-QBD4	TWO-WAY DISCONNECTOR		
F	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER		

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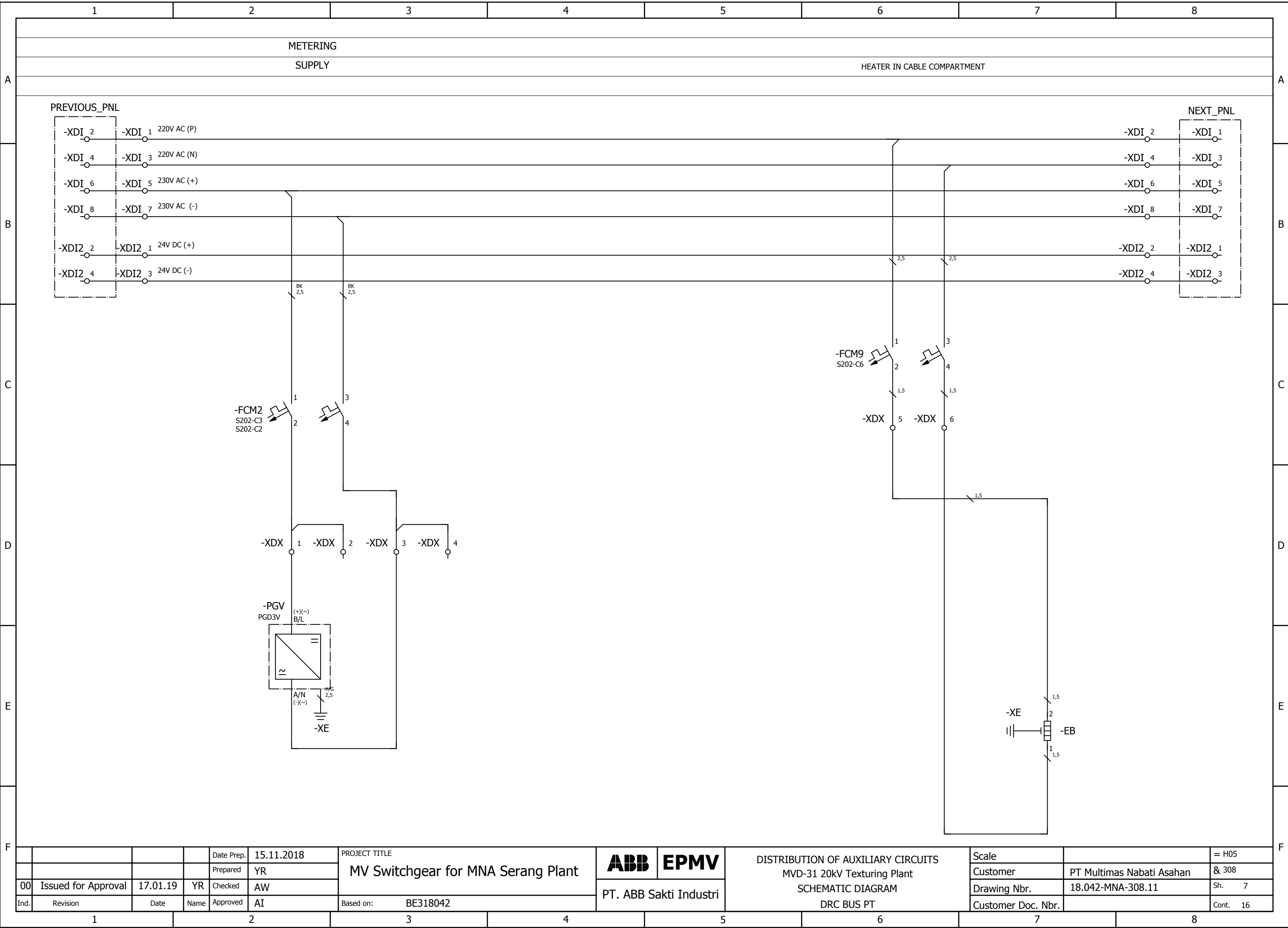
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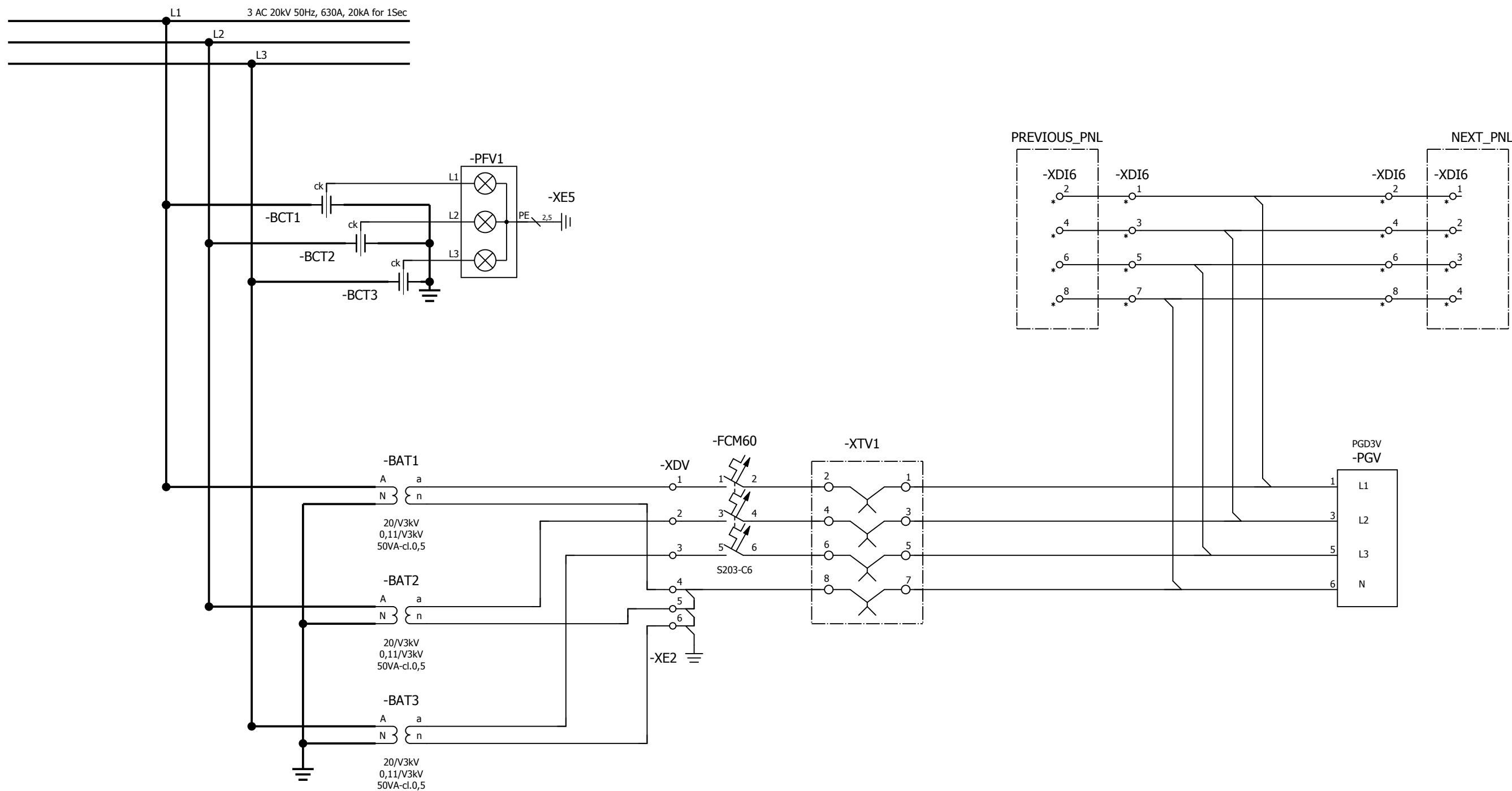
A	-TFA	ACTIVE POWER TRANSDUCERS	-XD11	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS	A
	-TFC	CURRENT TRANSDUCERS		-XD12	
B	-TFF	FREQUENCY TRANSDUCERS	-XD13	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION	B
	-TFJ	ACTIVE ENERGY TRANSDUCERS		-XD14	
	-TFK	REACTIVE ENERGY TRANSDUCERS	-XD16	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS	
	-TFM	MULTIFUNCTION TRANSDUCERS	-XD17	TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS	
	-TFP	POWER-FACTOR TRANSDUCERS	-XD18	TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS	
	-TFQ	REACTIVE POWER TRANSDUCERS	-XD19	TERMINAL BLOCK FOR SPECIAL USE	
	-TFS	SIGNAL CONVERTERS	-XDJ	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD	
	-TFV	VOLTAGE TRANSDUCERS	-XDJ1	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS	
	-WA	BUSBARS	-XDM	SEALABLE TERMINAL BLOCK FOR MEASUREMENT	
	C	-WBC	POWER CABLES	-XDS	
-WE		EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.	-XDT	TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB	
-WGC		CONTROL CABLES	-XDU	CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK	
-WGS		COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)	-XDU1	TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK	
-WH		OPTICAL FIBRES	-XDV	TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS	
-XDA		TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS	-XDV1	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1	
-XDA1		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1	-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2	
-XDA2		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2	-XDV3	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3	
-XDA3		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3	-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR	
D		-XDA5	CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER	-XDX	SUPPORT TERMINAL BLOCK
	-XDA8	CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS	-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE	
	-XDB	CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS	-XE2	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE	
	-XDB1	CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)	-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE	
	-XDB2	CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER	-XE6	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE	
	-XDB10...-XDB89	CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS	-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE	
	-XDB7	CONNECTOR FOR TRUCK POSITION CONTACTS	-XE8	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE	
	-XDB91	CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR	-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT	
	-XDB92	CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR			
	E	-XDB93	CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR		
-XDB95		CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET			
-XDC		CUSTOMER TERMINAL BLOCK			
-XDC1		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER			
-XDC2		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR			
-XDC3		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT			
-XDC4		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS			
-XDC5		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER			
-XDE		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE			
F		-XDE1	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE		
	-XDE2	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2			
	-XDE3	TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2			
	-XDF	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS			
	-XDF1	CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR			
	-XDF2	CONNECTOR FOR REAR VENTILATOR CIRCUITS			
	-XDF3	CONNECTOR FOR FRONT VENTILATOR CIRCUITS			
	-XDH	TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)			
	-XDI	TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)			

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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	MAIN CIRCUITS MVD-31 20kV Texturing Plant SCHEMATIC DIAGRAM DRC BUS PT	Scale		= H05	
				Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan	& 308
00	Issued for Approval	17.01.19	YR	Checked	AW		Drawing Nbr.	18.042-MNA-308.11			Sh. 8		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	Customer Doc. Nbr.					Cont. 16	

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1

2

3

4

5

6

7

8

A

B

C

D

E

F

490

580 mm

1

9.000

0.00

0.00

0.00

0

0

0

0

0.00

2

PTT

CUT OUT DIMENSION (in mm)

DIGITAL VOLTMETER

93

44

TEST BLOCK

5

80

14

17

94

				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	LV DOOR LAYOUT MVD-31 20kV Texturing Plant SCHEMATIC DIAGRAM DRC BUS PT	Scale		= H05
				Prepared	YR	MV Switchgear for MNA Serang Plant					Customer	PT Multimas Nabati Asahan	& 308
00	Issued for Approval	17.01.19	YR	Checked	AW			Based on:	BE318042		Drawing Nbr.	18.042-MNA-308.11	Sh. 9
Ind.	Revision	Date	Name	Approved	AI	Customer Doc. Nbr.					Cont. 16		

1

2

3

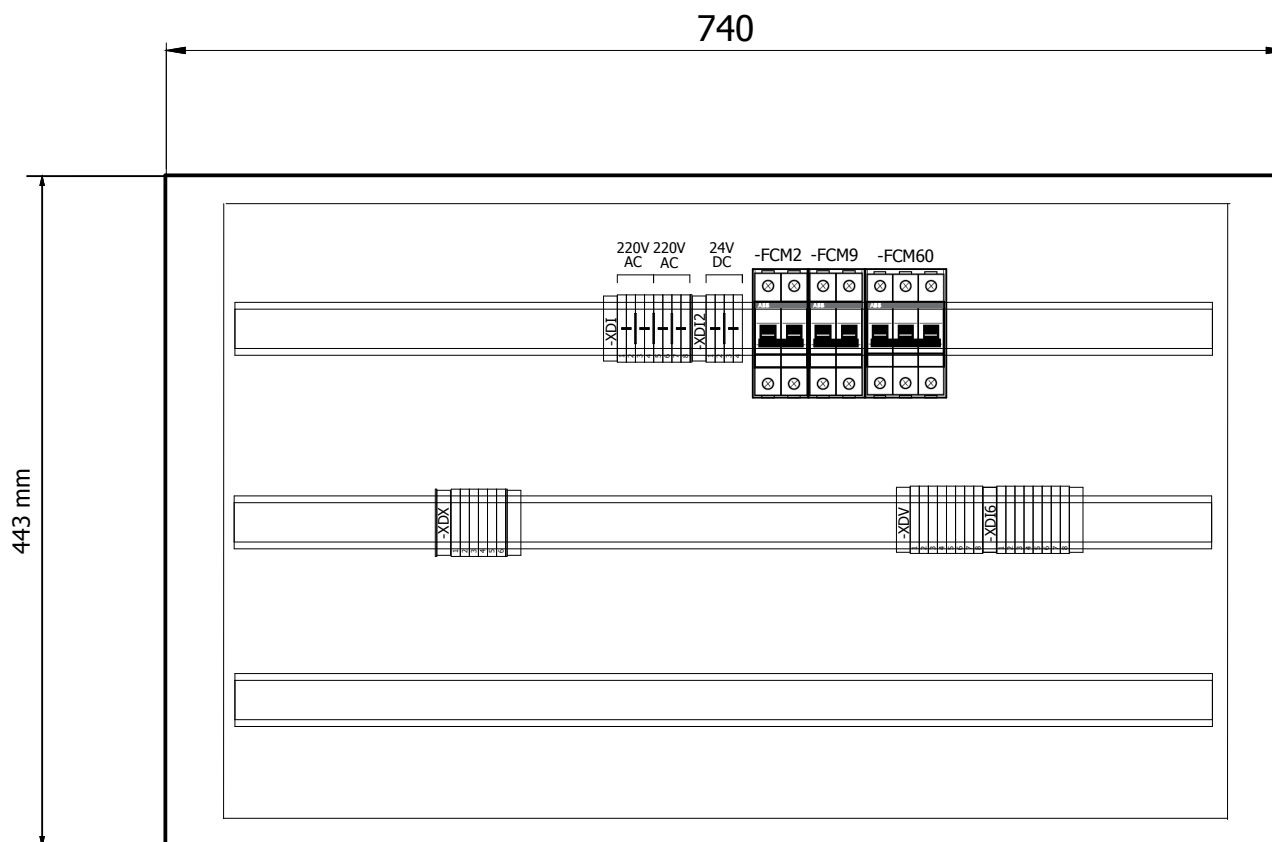
4

5

6

7

8

[illegible]

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	LV BOX LAYOUT MVD-31 20kV Texturing Plant SCHEMATIC DIAGRAM DRC BUS PT	Scale		= H05		
				Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan	& 308	
00	Issued for Approval	17.01.19	YR	Checked	AW		Drawing Nbr.	18.042-MNA-308.11			Sh.	10		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	Customer Doc. Nbr.					Cont.	16	

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1		2		3		4		5		6		7		8	
Summarized parts list															
IDABB_F02_005-IDABB															
Carry over: 0.00															
TAG ID		DESCRIPTION				MANUFACTUR/BRAND		PART NUMBER		QUANTITY		REMARKS			
-EB		HEATER 230VAC, 50W				SANYANG		CDXN010033P0009		1		/7.7:E			
-FCM2		MCB S202-C2 MCB. 2P, 2A				ABB		2CDS252001R0024		1		/7.2:C			
-FCM9		MCB S202-C6 MCB. 2P, 6A				ABB		2CDS252001R0064		1		/7.6:C			
-FCM60		MCB S203-C6 MCB. 3P, 6A				ABB		2CDS253001R0064		1		/8.4:C			
-PGV		Digital AC Voltmeter Model : DPM PGD3V, 48 X 96 MM				RISHABH		DPM PGD3V		1		/7.2:D			
-XTV1		Voltage Test Block Rated Current : 10A, 250V; Mounting : Flush Mounting				YONGSUNG		YSPTT-04C		1		/8.5:C			



EPMV

PART LIST
MVD-31 20kV Texturing Plant
SCHEMATIC DIAGRAM
DRC BUS PT

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]